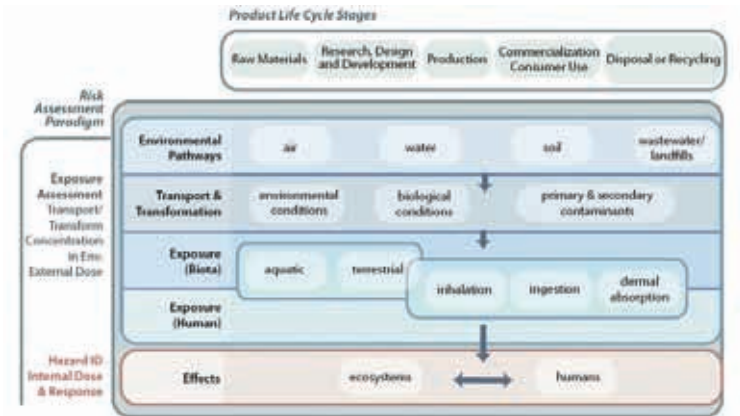




The risk assessment paradigm integrated with nanomaterial life cycle stages.

and the presence of co-contaminants. The transformed nanomaterials, in turn, may modify atmospheric, soil, or water chemistry. These transformations may alter the form of the nanomaterials to which humans and ecosystems are exposed and which are transported through the environment. Biological or environmental systems may be exposed to these dispersed nanomaterials and in ways that could perturb human health or adversely impact the environment.

Health

As nanotechnology is an emerging field, there is great debate regarding to what extent nanotechnology will benefit or pose risks for human health. Although nanoparticles owe all its distinguishing properties to its small size, this very nature makes them a likely suspect for entering the human body unintentionally. The behavior of nanoparticles is a function of their size, shape and surface reactivity with the surrounding tissue. Nano-toxicology can occur in multiple ways:

- ▶ They could cause overload on phagocytes, cells that ingest and destroy foreign matter, thereby triggering stress reactions that lead to inflammation and weaken the body's defense against other pathogens.
- ▶ The large surface of nanoparticles, on exposure to tissue and fluids, will immediately adsorb onto their surface some of the macromolecules they encounter. This may, for instance, affect the regulatory mechanisms of enzymes and other proteins.

The large number of variables influencing toxicity means that it is dif-